

FR-018 M2 — URL Acquisition + Thin Discovery Ingress

FR018 — `sources/optional/m2.md`

Masterclass source package snapshot

Functional Requirement	FR-018
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M2)** — **GO WITH REVISED SCOPE
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python `scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

Contents

FR-018 M2 — URL Acquisition + Thin Discovery Ingress

Status: Complete (M2) — GO WITH REVISED SCOPE

Date: 2026-08-07

Phase: Horizon 1B (lead FR)

Preceding: [M0](#) (Accepted); [M1](#) (Accepted); [ADR-010](#)

Scope (M2): Executable URL acquisition for owner-supplied SEEK / LinkedIn / Indeed job URLs; thin `ThinDiscoveryIngress`; `cic opportunity discover`; offline CI fixtures. **No** Playwright, email, feeds, scheduling, or Horizon 1A redesign.

1. Executive summary

M2 delivers the first executable FR-018 path:

```
owner URL → UrlAcquisitionAdapter → ThinDiscoveryIngress
           → ApplicationWorkflowRunner → Opportunity / review (frozen 1A)
```

Offline fixture tests prove `acquire` → `provenance` → `definite idempotency` → `runner handoff`. Live HTTP against a sample SEEK URL from this environment failed closed with `network_failure` (no Playwright escalation). **Supported sources are implemented and fixture-validated**; live board availability is **environment / platform dependent** and must be validated by the owner on real public URLs.

Verdict: GO WITH REVISED SCOPE — ship SEEK/LinkedIn/Indeed *attempt* path + clear unsupported/blocked failures; do not claim unconditional live success for LinkedIn/Indeed (or SEEK) without owner live confirmation; no scrape escalation.

2. Implementation summary

Component	Location
HTTP boundary	<code>discovery/http.py</code> (<code>HttpFetchClient</code> , <code>UrllibHttpClient</code> , <code>FakeHttpClient</code>)
URL classify / normalise	<code>discovery/url_support.py</code>
HTML extract (narrow)	<code>discovery/extract.py</code>
<code>UrlAcquisitionAdapter</code> / <code>StaticAcquisitionAdapter</code>	<code>discovery/url_adapter.py</code>
Definite idempotency	<code>discovery/idempotency.py</code> (reuses FR-009 <code>classify_pair</code>)
<code>ThinDiscoveryIngress</code>	<code>discovery/thin_ingress.py</code>
CLI	<code>cic opportunity discover <url></code>

3. Supported sources

Platform	Locator requirement	Fixture validated	Live (this env)
SEEK	<code>/job/<id></code>	Yes	Network failure on sample URL
LinkedIn	<code>/jobs/view/<id></code> or <code>currentJobId</code>	Yes (HTML fixture)	Not live-tested here; often login-walled
Indeed	<code>jk=</code> job key	Yes (HTML fixture)	Not live-tested here
Other hosts	—	Fail <code>unsupported_source</code>	—

Unsupported company career pages fail closed — no clever generic scrape.

4. URL fetch / extraction design

- Injectable `HttpFetchClient` — CI uses `FakeHttpClient` + saved HTML.
- Live: stdlib `UrllibHttpClient` (no new HTTP dependency).
- Timeout → `network_failure` ; HTTP errors mapped; blocked/login-wall heuristics → fail closed.
- Extraction: strip script/style; prefer board description regions; min length; no LLM.

WHY injectable fetch: Offline deterministic CI.

Alternative: Live HTTP in tests. **Rejected:** flaky / non-CI.

Principle: Validate first; adapter isolation.

Interview: Anti-corruption at the network boundary.

5. Provenance behaviour

After acquire, ingress calls `assert_url_acquisition_provenance` requiring `source_kind=url`, `source_url`, `source_identifier`, `acquired_at`, non-empty `raw_content`. Failures → `partial_metadata` / `malformed_content`.

6. Canonicalisation / identity

- `derive_source_facets` (existing) for platform / job id / canonical URL.
 - Tracking query stripped for fetch (`utm_*`); job keys (`jk`, `currentJobId`) kept.
 - `source_identifier` = canonical URL.
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7. Idempotency / duplicates

- Pre-fetch skip when URL facets form a **definite** FR-009 match.
 - Post-acquire definite check again.
 - Fingerprint-only → **not** definite → does not auto-skip.
 - Skip returns `matched_opportunity_id`; never merge/delete.
 - `--force` bypasses skip.
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8. DiscoveryIngress behaviour

`ThinDiscoveryIngress.discover` :

1. validate / classify supported URL
2. optional definite skip
3. `UrlAcquisitionAdapter.acquire`
4. provenance assert
5. optional definite skip
6. `runner.start(StaticAcquisitionAdapter(result))` — **one** HTTP fetch

7. return `DiscoveryOutcome` item

No discovery catalogue writes.

9. CLI design

Command: `cic opportunity discover <url>`

WHY under `opportunity`: Discovery creates Opportunities into the existing SoT; keeps one owner namespace vs a new `discovery` tree.

Alternative: `cic discovery url`. **Rejected:** extra hierarchy for one verb.

Flags: `--dir`, `--checkpoint-dir`, `--profile`, `--force`, `--offline-fixtures`.

Outcomes printed: `ACQUIRED` / `SKIPPED_ALREADY_REPRESENTED` / `FAILED <kind>`.

10. Failure semantics

Case	Kind / status
Invalid URL	<code>invalid_url</code>
Unsupported host / incomplete locator	<code>unsupported_source</code>
Timeout / DNS / HTTP	<code>network_failure</code>
Blocked / login wall	<code>adapter_failure</code>
Empty / short HTML	<code>malformed_content</code>
Missing provenance	<code>partial_metadata</code>
Definite duplicate	<code>skipped</code> + <code>definite_identity_match</code>
Runner exception	<code>runner_failure</code>

11. Test results

Suite	Result
tests/unit/discovery/	35 passed
FR-008 acquisition/runner/pre-review subset	42 passed
FR-009 detection subset	included in 42

Coverage includes: supported success, invalid/unsupported, timeout, HTTP error, insufficient content, blocked page, provenance, platform ids, definite skip, fingerprint non-skip, single fetch, runner failure, no discovery catalogue.

12. Manual validation

Scenario	Result (engineering env)
SEEK sample URL live fetch	FAILED <code>network_failure</code> (fail closed; no Playwright)
Fixture SEEK / LinkedIn / Indeed HTML	PASS (unit)
Duplicate URL twice (offline ingress)	PASS skip
Unsupported <code>example.com</code>	PASS <code>unsupported_source</code>

Owner action: Run `cic opportunity discover <real-public-url>` on a reachable network; if boards block plain HTTP, treat as known limitation — paste/export remain available. Do **not** add Playwright in M3 solely to force those boards.

13. Horizon 1A regression

No changes to runner graph, Opportunity SoT semantics, FR-009 detection rules, truth, agents, or submission. Orchestration regression subset green.

14. Risks / limitations

- Live board HTML often JS/login gated → ordinary HTTP may fail (documented).
- Title/company extraction is best-effort; warnings when missing.
- Linear Opportunity scan for idempotency (acceptable single-user).

- CLI live path still needs `OPENAI_API_KEY` unless `--offline-fixtures`.

15. Learning outcomes

Decision	WHY	Rejected alternative	Principle	Interview
Static adapter after acquire	One fetch	Double acquire in runner	Thin interfaces	Cache at boundary
FR-009 definite only	No false skip	Fingerprint auto-skip	Reuse SoT rules	Idempotency ≠ fuzzy dedupe
No Playwright in M2	Lawful / maintainable	Scrape to “make tests pass”	No scrape theatre	Fail closed honestly
<code>opportunity discover</code> CLI	Fits SoT verb	New command tree	Scope control	Smallest owner UX

16. Documentation updated

This report; changelog § 1.118; roadmap; functional spec; domain; testing; implementation notes; phase history; AGENTS/README/repo guide; ADR-010 note.

17. M2 Go / No-Go

GO WITH REVISED SCOPE

- Proceed to M3 only for ops/batch polish **after** owner live validation on at least one real URL, **or** accept paste/export fallback when boards block HTTP.
- Do **not** introduce Playwright as the M3 default.
- Email/feeds remain later milestones.

Post-M2 live validation (2026-08-07): Complete — [fr018_post_m2_live_url_validation.md](#). SEEK **SUPPORTED**; LinkedIn partial; Indeed blocked. Recommend **M3 = A (SEEK-first URL ops)**; M3 not started in that task.

18. Final repository status

Item	Status
Horizon 1A	Frozen
FR-018 M0–M1	Accepted
FR-018 M2	Complete — GO WITH REVISED SCOPE
M3	Not started

M2 COMPLETE — GO WITH REVISED SCOPE